Anway Pimpalkar

PhD Student in Engineering Sciences at Harvard University

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EDUCATION

2025 – **Harvard University** Cambridge, MA
PhD, Engineering Sciences: Materials Science & Mechanical Engineering
Secondary Field in Mind, Brain, and Behavior
Advisors: Katia Bertoldi, PhD and Patrick Slade, PhD

2023 – 2025 **Johns Hopkins University** Baltimore, MD
MSE, Biomedical Engineering
Thesis: *Bridging Perception & Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Feedback*
Advisor: Jeremy D. Brown, PhD

2019 – 2023 **College of Engineering, Pune** India
BTech, Electronics and Telecommunication Engineering

AWARDS & FELLOWSHIPS

2025 – 2031 **Prize Fellowship**, Harvard John A. Paulson School of Engineering and Applied Sciences
$281,796 (6 years of tuition, 2 years of stipend)

2025 – 2028 **National Defense Science & Engineering Graduate Fellowship (NDSEG)**
$277,484 (3 years of tuition, 3 years of stipend)

2025 – 2031 **GEM Associate Fellowship**
Full PhD tuition and fees sponsored by Harvard University

2025 **Best Student Paper Finalist**, International Conference on Rehabilitation Robotics [C2]

2025 **NIH NIA Start-Up Challenge Finalist**
$10,000 grant and 5-month accelerator with 1:1 mentorship and SBIR/STTR funding training

2023 **Best Paper Award**, International Conference on Biomedical Engineering Science & Technology [C1]

2022 **Mitacs Globalink Research Award**
CA $9,915

2021 **Best Project Award**, IEEE National Project Competition, India [O1]

RESEARCH EXPERIENCE

Oct 2025 – **Harvard University** Cambridge, MA
Doctoral Research Fellow • Bertoldi Group (Katia Bertoldi, PhD), Ability Lab (Patrick Slade, PhD)
Designing adaptive devices to augment human cognition, behavior, and performance.

Aug 2026 – **Boston Children's Hospital** Boston, MA
Affiliated Doctoral Researcher • Laboratories of Cognitive Neuroscience (Anne Arnett, PhD)
Investigating the neural correlates of optimal stimulation in children with ADHD.

Oct 2025 – Jun 2026 **Massachusetts Institute of Technology** Cambridge, MA
Doctoral Research Fellow • Laboratory for Translational Engineering (Giovanni Traverso, MD, PhD)
Developed ingestible stress sensing and solely mechanically tuned drug delivery platforms.

Jan 2024 – Jun 2025 **Johns Hopkins University** Baltimore, MD
Graduate Research Assistant • Haptics and Medical Robotics Lab (Jeremy D. Brown, PhD)
Developed cross-modal matching methods to quantify and balance visual and haptic perception, on robotic and smartphone platforms [C2, C3, C5, A7]. Evaluated perceptually calibrated visual-haptic feedback for finger individuation training in a virtual precision grip task, toward post-stroke dexterity rehabilitation [J3, A6]. Designed soft pneumatic and 3D-printed interfaces for multimodal tactile feedback [J4, A3, A4].

May 2024 – Aug 2024 **Harvard University** Cambridge, MA
Research Fellow • Biohybrid Organs and Neuroprosthetics Lab (Shriya Srinivasan, PhD)
Investigated peripheral nerve stimulation to modulate tumor metastasis. Explored a surgical approach for treating stress urinary incontinence.

Jan 2023 – Aug 2023 **IIT Bombay** Mumbai, India
Research Intern • Human Motor Neurophysiology and Neuromodulation Lab (Nivethida T, PhD)
Applied machine learning to quantify the prospective component of the sense of agency.

Aug 2022 – Aug 2023 **College of Engineering, Pune** India
Undergraduate Researcher • Department of Applied Sciences (Mahesh H. Shindikar, PhD)
Benchmarked vanilla, residual, and dense 2D U-Net architectures for skull stripping of T1-weighted MRI head scans [C1, A1].

May 2022 – Aug 2022 **Dalhousie University** Truro, Canada
Mitacs Globalink Research Intern • Agricultural Mechanized Systems Group (Travis J. Esau, PhD)
Contributed to automating industrial wild blueberry harvesters to reduce operator cognitive load.

Jun 2021 – Sep 2021 **Queliz Lifetech** Pune, India
Research Intern, Electronics Subsystem
Developed the electronics subsystem of a point-of-care device for finger dexterity rehabilitation in acute burn patients.

Journal Publications

J4 **Pimpalkar A**, Brown JD. Soft 3D-printed interface for co-located transient and continuous tactile stimuli. *In prep.*

J3 **Pimpalkar A**, West AM, Rai D, Xu J, Brown JD. Perceptually calibrated haptic feedback provides limited benefit beyond vision in a virtual precision grip task. 2026. *In review*.

J2 Harris C, **Pimpalkar A**, Aggarwal A, Yang J, Chen X, Schmidgall S, Rapuri S, Greenstein JL, Taylor CO, Stevens RD. Preoperative risk prediction of major cardiovascular events in noncardiac surgery using the 12-lead electrocardiogram: explainable deep learning approach. *British Journal of Anaesthesia*. 2025 Nov;135(5):1161–1171. 10.1016/j.bja.2025.07.085.

J1 **Pimpalkar A**, Slepyan A, Thakor NV. Vibrations at first contact encode object stiffness before grasp completion. *IEEE Sensors Letters*. 2025 Aug;9(8):1–4. 10.1109/lsens.2025.3584860.

Conference Presentations

† presenter

Full-Length Papers

C5 West AM†, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. Cross-modal matching is not bidirectional: Implications for human multisensory feedback. *IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics* (*BioRob*). 2026 Aug. p. 1222–1229. 10.1109/biorob66782.2026.11681322. (Oral)

C4 Salgado AS, Bartlett JS, **Pimpalkar A**, Brown JD, Reed KB†. Teaching at the teachable moment using asynchronous videos combined with live interactions. *ASEE Annual Conference & Exposition*. 2026 Jun. 10.18260/1-2--60737. (Oral)

C3 West AM†, **Pimpalkar A**, Xu J, Brown JD. SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments. *International IEEE/EMBS Conference on Neural Engineering* (*NER*). 2025 Nov. p. 85–90. 10.1109/ner61569.2025.11589033. (Spotlight poster)

C2 **Pimpalkar A**†, West AM, Xu J, Brown JD. Optimizing cross-modal matching for multimodal motor rehabilitation. *International Conference on Rehabilitation Robotics* (*ICORR*). 2025 May. p. 559–566. 10.1109/icorr66766.2025.11063112. (Oral) **Best Student Paper Award Finalist.**

C1 **Pimpalkar A**†, Patole RK, Kamble KD, Shindikar MH. Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans. *International Conference on Biomedical Engineering Science and Technology* (*ICBEST*). 2023 Feb. p. 131–142. 10.1007/978-3-031-54547-4_11. (Oral) **Best Paper Award.**

Abstracts and Short Papers

A7 West AM†, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. Cross-modal matching is not bidirectionally invertible: Implications for multisensory representation. *Neural Control of Movement Meeting*. 2026 Apr. (Poster)

A6 **Pimpalkar A**†, Rai D, Bartels UJ, Xu J, Brown JD. Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task. *Biomedical Engineering Society Annual Meeting* (*BMES*). 2024 Oct. (Oral)

A5 Song H†, Patterson A, Kang J, Cheng J, Keuthan CJ, Berlinicke C, Weiler C, Xu K, **Pimpalkar A**, Lane N, Shrekenhamer D, Gehlbach PL, Spicer J, Billings S, Zack DJ, Boctor E. Towards visual function restoration through photoacoustic stimulation. *Investigative Ophthalmology & Visual Science ARVO Annual Meeting*. 2024 May. (Oral)

A4 **Pimpalkar A**†, Ameta P†, Dalia A, Brown JD. Pneumatactor arrays for high frequency vibrotactile feedback. *IEEE Haptics Symposium*. 2024 Apr. (Live demo & poster)

A3 **Pimpalkar A**†, Ameta P†, Dalia A†, Brown JD. Pneumatactor arrays for high frequency vibrotactile feedback. *US Senate AI Caucus Robotics Demo Day*. 2024 Apr. (Live demo)

A2 Harris C†, **Pimpalkar A**, Aggarwal A, Yang P, Chen X, Taylor CO, Greenstein J, Stevens RD. Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms. *Johns Hopkins Medicine & Engineering Research Retreat*. 2024 Feb. (Poster)

A1 **Pimpalkar A**†, Patole R, Kamble K, Shindikar M. Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans. *No Garland Neuroscience Conference*. 2023 Feb. (Oral)

Non-Peer-Reviewed Articles

O1 **Pimpalkar AS**, Niture DV. Towards contactless elevators with TinyML using CNN-based person detection and keyword spotting. *arXiv*. 2024 May. arXiv:2405.13051. **Best Project Award.**

Inventions Disclosed

P1 Deep learning models for predicting major adverse cardiovascular events in noncardiac surgery. 2026 Aug. *Disclosed to Johns Hopkins Technology Ventures*.

Teaching Experience

2026 – 2027 **Engineering Design (ES 100)** Harvard University
Teaching Fellow
Mentoring 6 student teams through year-long, industry-proposed capstone projects using agile design, spanning human-robot interaction, lightweight structures, sustainability, complex-environment navigation, and surgical robotics.

2024 **Haptic Interface Design for Human-Robot Interaction (530.691)** Johns Hopkins University
Class Researcher
Helped pilot an interactive video platform that anchors student questions and instructor responses to lecture timestamps, bringing live interaction to a self-paced hardware design course; co-authored the resulting ASEE 2026 paper. [C4]

2024 **Principles of the Design of Biomedical Instrumentation (580.771)** Johns Hopkins University
Teaching Assistant
Mentored labs on ECG, EMG, and EEG circuit design, and 24 student projects ranging from accessible gaming to wearables.

2024, 2025 **Mechatronics (530.421)** Johns Hopkins University
Teaching Assistant
Solved a 3-year-old challenge of real-time location polling for autonomous air hockey robots using computer vision (ArUco tags) and ZigBee mesh networking; led labs on building autonomous robots. Article / GitHub

Entrepreneurial Experience

Nov 2024 – Sep 2025 **Re-Kinesis** Baltimore, MD
CEO and Co-Founder • NIH NIA Start-Up Challenge Finalist
Led a medical device venture building a tactile insole that reconstructs gait kinematics for mobility assessment in aging populations. Article / NIH Announcement

Mentoring

Undergraduate Students

Oct 2026 – **Arya Bosworth**, *Undergraduate Researcher from Harvard University*
Mechanics of elastic snap-through structures.

Oct 2026 – **Suhani Sharma**, *Undergraduate Researcher from Harvard University, Electrical Engineering*
Actively tunable structures.

Jun 2026 – Aug 2026 **Jocelyn Hwang**, *Summer Intern from MIT, Electrical Engineering and Computer Science*
Multiplexing electronics for multi-electrode biosensing.

Jul 2026 – Aug 2026 **George Sealy**, *Summer Intern from University of Cambridge, Materials Science*
Mechanics of elastic structures.

Jan 2026 – Jul 2026 **Fu Xing Chen**, *Undergraduate Researcher from MIT, Mechanical Engineering*
Mechanics and characterization of viscoelastic actuators.

Invited Presentations and Lectures

2025 **IIT Bombay**, Human Motor Neurophysiology & Neuromodulation Lab Mumbai, India
Invited Talk • "Generating, Calibrating, & Evaluating Multimodal Sensory Feedback"

2024 **Johns Hopkins University**, Neural & Rehabilitation Engineering Baltimore, MD
Guest Lecture • "The Role of Haptic Augmentation in Rehabilitation Engineering"

Academic Service

2024 **IEEE Haptics Symposium**, *Student Volunteer*

2022 **Canadian Society of Bioengineering Annual Meeting**, *Student Volunteer*

In Media

2025 Sep 17 — *Johns Hopkins University Hub*
"AI fares better than doctors at predicting deadly complications after surgery"

2025 May 9 — *Johns Hopkins University Hub*
"A game-changing gait lab in a shoe"

2025 Jan 2 — *New Scientist Magazine*
"Delicate robot hands know just how hard to squeeze"

2024 May 16 — *Johns Hopkins Whiting School of Engineering*
"Robotics teams shoot it out in JHockey tournament"

2022 Jan 29 — *Arduino Blog*
"This contactless system combines embedded ML and sensors to improve elevator safety"

Leadership and Teamwork

2024 – 2025 **Johns Hopkins Gupta-Klinsky India Institute** Washington, DC
Operations Team
Managed day-of operations and logistics for the inaugural Hopkins-India Conference.

2024 – 2025 **Johns Hopkins Biomedical Engineering** Baltimore, MD
Master's Program Representative
Represented the department by invitation at information sessions and conferences.

2019 – 2021 **COEP Impressions Cultural Festival** India
Head of Events (2021), Volunteer
Led 28 events (10k attendees, ₹2M budget) with a 126-member team, directly managing 15 members alongside 2 co-heads.

2020 – 2022 **Rowing Team of College of Engineering, Pune** India
1X, 2X, 4X Sculler
Won races and an institutional award for skill and spirit; represented Pune City at regionals.